E-Commerce Sales and Customer Intelligence – Exploratory Data Analysis of the Superstore Dataset using Python, Pandas, Matplotlib and Seaborn

A PROJECT REPORT

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Submitted to
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BONAFIDE CERTIFICATE

Certified that this project report **"E-Commerce Sales and Customer Intelligence – Exploratory Data Analysis of the Superstore Dataset using Python, Pandas, Matplotlib and Seaborn"** is the bonafide work of **"Bhumi Jindal (26MCD10071)"** who carried out the project work under my/our supervision. The content and findings presented herein are authentic and reflect the original work conducted during the course of this project.

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ABSTRACT

Retail and e-commerce businesses record every order they receive, but thousands of raw transaction rows cannot by themselves tell a manager which months, products or customer groups drive revenue and profit. This project, **E-Commerce Sales and Customer Intelligence**, is an exploratory data analysis (EDA) of the well-known *Sample – Superstore* dataset, which contains 9,994 order line items placed between January 2014 and December 2017. The analysis is carried out in a Jupyter Notebook using Python, pandas, NumPy, Matplotlib and Seaborn.

The dataset is first loaded, profiled and validated: it has 21 columns, no missing values and no duplicate rows. Order and ship dates are converted to date-time type, and three time features (order month, order year and order weekday) are derived. Sales and profit are then aggregated by **month**, **product category**, **product sub-category** and **customer segment**, and presented through line charts, donut charts and horizontal bar charts. A sales-to-profit ratio is also calculated for each segment.

The analysis shows total sales of about **$2.30 million** and total profit of about **$0.29 million**, an overall profit margin of **12.5%**. Sales peak in **November** ($352,461) and are lowest in **February** ($59,751). **Technology** is the largest category by both sales (36.4%) and profit (50.8%), whereas **Furniture** contributes 32.3% of sales but only 6.4% of profit. **Phones** and **Chairs** are the best-selling sub-categories and **Copiers** the most profitable, while **Tables**, **Bookcases** and **Supplies** make a loss. The **Consumer** segment brings 50.6% of sales, but **Home Office** earns the best margin (14.0%).

The work is published on GitHub together with the dataset and the charts. This report describes the scope, background, design, implementation, results and future scope of the project, and includes the charts produced by the analysis.

Keywords: Exploratory Data Analysis, E-Commerce, Sales Analysis, Customer Segmentation, Python, pandas, Matplotlib, Seaborn, Jupyter Notebook.

List of Abbreviations

Abbreviation	Definition
EDA	Exploratory Data Analysis
CSV	Comma-Separated Values
KPI	Key Performance Indicator
BI	Business Intelligence
IDE	Integrated Development Environment
ID	Identifier
YoY	Year over Year
RFM	Recency, Frequency, Monetary (customer scoring method)
CLV	Customer Lifetime Value
ML	Machine Learning
UI	User Interface

CHAPTER 1.
INTRODUCTION

Online and offline retailers record every order they receive: who bought, what was bought, where it was shipped, how much was paid, what discount was given and how much profit remained. When thousands of such records accumulate, the data can no longer be understood line by line, yet it holds the answers to questions that managers ask every day: Which months are the busiest? Which product lines make money and which lose it? Which customer groups are worth the most? Exploratory data analysis (EDA) answers such questions by summarising the data with statistics and visualisations.

E-Commerce Sales and Customer Intelligence applies EDA to the Sample – Superstore dataset, a retail transaction file of 9,994 order lines covering four years. The project is deliberately built with open-source Python libraries inside a Jupyter Notebook, so that every step – loading, cleaning, feature creation, aggregation and visualisation – is visible, reproducible and can be re-run on any other sales file with the same structure. It therefore also serves as a practical demonstration of how raw sales records are turned into business insight.

1.1. Project Scope and Project Planning:

The scope of the project is the exploratory analysis of the Superstore sales data using Python. The analysis covers: (i) loading and profiling of the dataset; (ii) validation of data quality (missing values, duplicates and data types); (iii) creation of time-based features from the order date; (iv) the monthly sales trend; (v) sales by product category and sub-category; (vi) profit by month, category and sub-category; and (vii) sales, profit and the sales-to-profit ratio for each customer segment. Forecasting, machine-learning models, interactive dashboards, regional analysis and customer-level scoring are outside the present scope and are listed under future work.

1.2. Client Identification and Recognition of need: -

The primary clients of the analysis are the store managers and the sales and marketing managers of a retail or e-commerce company, who decide what to stock, when to run promotions and which customers to target. Secondary stakeholders are inventory and procurement planners, who need to know which products move and which do not, and the finance team, who track profitability. Their typical needs are: a clear view of how sales change through the year; knowing which categories and sub-categories earn the most revenue and profit; spotting products that sell well but lose money; comparing customer segments; and charts that need no technical background to read. These needs were converted into the objectives in Section 2.6.

1.3. Recognition and Knowledge of relevant contemporary issues: -

E-commerce has grown rapidly, and businesses of every size are expected to make decisions from data rather than intuition. Several contemporary issues are relevant to this project. **Margin pressure:** high sales do not guarantee profit, because discounts, shipping and product cost can turn large orders into losses; in this dataset a single order line loses as much as $6,600. **Seasonality:** demand is uneven through the year, which affects inventory, staffing and cash flow. **Customer segmentation:** treating all customers alike wastes marketing effort, so businesses compare groups such as consumers, corporates and home offices. **Data quality and reproducibility:** conclusions are reliable only if the data is checked and the steps can be repeated. **Privacy:** real sales data contains personal details such as customer names, which must be protected; the Superstore file used here is a public sample dataset.

1.4. Task Identification: -

- **Data task:** load `Sample - Superstore.csv` with the correct encoding and inspect its structure, types and statistics.
- **Quality and feature tasks:** check missing values and duplicates, convert the date columns and derive Order Month, Year and Weekday.
- **Sales and profit tasks:** aggregate both by month, category and sub-category; find the highest and lowest values and the loss-making sub-categories.
- **Customer task:** compare sales, profit and the sales-to-profit ratio across the three customer segments.

- **Visualisation tasks:** produce line, donut and bar charts with titles, labels and data values.
- **Quality and publication tasks:** verify the totals, save the charts, document the work and publish it on GitHub.

1.5. Technologies Used: -

Technology	Purpose in the project
Python	Core programming language used for the whole analysis.
pandas	Loading the CSV file, converting dates, grouping and aggregating sales and profit.
NumPy	Numerical support for pandas and for calculations.
Matplotlib	Base plotting library: line charts, donut (pie) charts, titles and labels.
Seaborn	Statistical plotting built on Matplotlib: line plots and horizontal bar plots with data labels.
Jupyter Notebook	Interactive environment that combines code, output and notes in one document.
Sample – Superstore.csv	Data source: 9,994 order lines, 21 columns, orders from January 2014 to December 2017.
Git, GitHub	Version control and hosting of the notebook, dataset, README and chart images.

1.6. Timeline of the project

The project was carried out in six phases, shown in Figure 1 – problem and data understanding, data cleaning and preparation, feature engineering, aggregation and analysis, visualisation and interpretation, and documentation with GitHub publication. Earlier steps were repeated whenever a later step revealed a data issue.

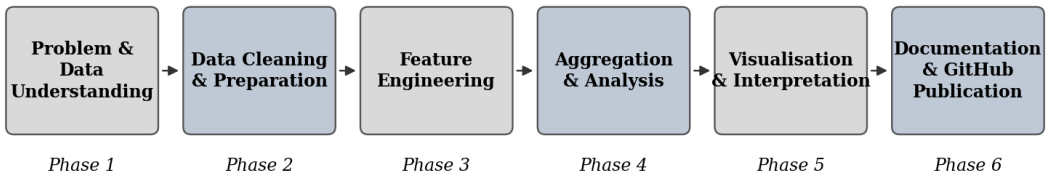


Figure 1: Development lifecycle of the project

CHAPTER 2.
LITERATURE REVIEW/BACKGROUND STUDY

2.1 Timeline of the reported problem:

Business reporting has moved from paper ledgers and manual tallies to spreadsheets (from the 1980s), then to relational databases, data warehouses and reporting tools (1990s and 2000s), and then to self-service business-intelligence software such as Tableau and Power BI. Since the 2010s, code-based analysis in Python and R has become common for analysts who need repeatable work. The idea of looking at data visually before building models was set out by John Tukey in his 1977 book on exploratory data analysis. At every stage the reported problems have been similar: data scattered across files, inconsistent or dirty data, totals that hide loss-making items, and a long delay between collecting data and acting on it.

2.2 Existing solutions:

Spreadsheets (e.g. Excel) are easy to start with and well known, but large files become slow, steps are hard to repeat exactly and mistakes in formulas are easy to miss. **BI tools** (e.g. Tableau, Power BI) offer interactive dashboards, but they are often licensed products and hide the calculations behind a graphical interface. **SQL**

reporting is efficient for aggregation but offers limited charting by itself. **Python notebooks** combine data handling, statistics and charts in one reproducible document, are free, and show every step of the analysis, which is the approach chosen for this project.

2.3 Bibliometric analysis:

A review of the textbooks, papers and official documentation listed in the References shows the following.

Key features

- The "split–apply–combine" pattern of pandas (`groupby` followed by `sum`) summarises transaction data at any level, such as month, category or segment.
- Matplotlib and Seaborn provide the standard chart types needed for EDA with only a few lines of code.
- Notebooks keep code, results and explanations together, which makes the analysis easy to review and share.

2.4 Review Summary:

The literature agrees that a sound EDA begins with checking data quality, summarises the data at meaningful levels and uses simple, well-labelled charts for each kind of question: lines for trends, bars for comparisons and part-to-whole charts for shares. It also warns that totals alone can mislead and that profit must be examined together with sales. This project adopts these practices and lists the deeper techniques (forecasting, clustering, dashboards) as future work.

2.5 Problem Definition:

To analyse the Sample – Superstore transaction dataset using Python in order to determine **how sales and profit vary across time, product category, product sub-category and customer segment**, to **identify the best- and worst-performing areas**, and to present the findings through clear visualisations that support business decisions.

2.6 Goals/ Objectives:

- To load, profile and validate the Superstore dataset using pandas.
- To create time-based features (order month, year and weekday) from the order date.
- To analyse sales and profit by month, category and sub-category and find the highest and lowest values.
- To compare customer segments by sales, profit and sales-to-profit ratio.
- To present the results as clear line, donut and bar charts and summarise the business insights.
- To publish the notebook, dataset and charts on GitHub so that the work can be reproduced.

CHAPTER 3.
DESIGN FLOW/PROCESS

The design follows the standard pipeline of a data-analysis project: acquire the data, prepare it, analyse it, visualise the results and communicate them. Each stage uses a small number of pandas, Matplotlib and Seaborn functions, so that the whole analysis stays easy to follow. The design process moved from identifying the features of the analysis, through constraints and selection, to the final workflow and implementation plan described below.

3.1 Evaluation & Selection of Features: -

Candidate features were evaluated for their usefulness to a sales manager and for the effort involved. Features marked "Included" form the implemented analysis; the others are deferred.

Feature	Reason	Priority	Decision
Data loading and profiling	Understand structure, types and statistics	High	Included
Data quality checks	Reliable results need clean data	High	Included

Feature	Reason	Priority	Decision
Time features (month, year, weekday)	Enable trend and seasonality analysis	High	Included
Sales analysis (month, category, sub-category)	Core revenue indicators	High	Included
Profit analysis (month, category, sub-category)	Find profitable and loss-making areas	High	Included
Segment analysis and sales-to-profit ratio	Customer intelligence	Medium	Included
Discount and regional analysis	Explain the causes of losses	Medium	Future
Interactive dashboard	Presentation for non-technical users	Low	Future
Forecasting and customer clustering	Advanced analytics	Low	Future

3.2 Design Constraints: -

- **Data:** a single historical CSV file (no live feed); it must be read with `latin-1` encoding; each row is an order line, not a whole order.
- **Technical:** the analysis must run in a standard Jupyter Notebook on an ordinary laptop with Python, pandas, NumPy, Matplotlib and Seaborn.
- **Analytical:** sales and profit are summed at the level of interest; the parts must add up to the overall total, and percentages must be taken from those totals.
- **Privacy:** the file contains customer names; it is a public sample dataset, but real data would have to be anonymised before sharing.
- **Resource and time:** a single-developer academic project, so static charts are used instead of a full dashboard.
- **Usability:** every chart must have a title, labelled axes and, where useful, data labels so that it can be read without explanation.

3.3 Analysis of Features and finalization subject to constraints:

Considering the constraints, all analyses that depend only on the columns Order Date, Category, Sub-Category, Segment, Sales and Profit were finalised, because they need no extra data and no extra tools. Discount and regional analysis were deferred because they enlarge the scope beyond the main sales and profit questions. An interactive dashboard and predictive models were deferred because they need additional tools and more time. Static Seaborn and Matplotlib charts were preferred to a dashboard because they can be embedded directly in this report and in the GitHub repository.

3.4 Design Flow:

Figure 2 shows the flow of the analysis. The Superstore CSV file is read into a pandas DataFrame; dates are converted and the data is checked for missing values and duplicates; time features are derived; the data is grouped and summed by month, category, sub-category and segment; the results are plotted with Matplotlib and Seaborn; and the notebook, data and charts are published on GitHub.

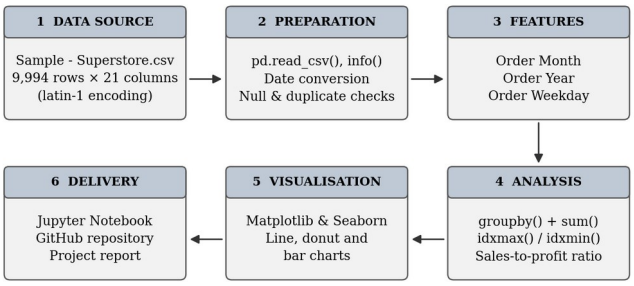


Figure 2: Workflow and architecture of the analysis pipeline

The structure of the input data is given in Table 3.1. The first 19 rows are the original columns of the file (some grouped), and the last three are derived during the analysis.

Table 3.1: Structure of the Superstore dataset

Column	Type	Description
Row ID	Integer	Unique row number of the order line
Order ID	Text	Identifier of the order (5,009 distinct orders)
Order Date	Date-time	Date on which the order was placed
Ship Date	Date-time	Date on which the order was shipped
Ship Mode	Text	Shipping method (e.g. Standard Class, Second Class)
Customer ID	Text	Identifier of the customer (793 distinct customers)
Customer Name	Text	Name of the customer
Segment	Text	Customer segment: Consumer, Corporate or Home Office
Country, City, State	Text	Delivery location
Postal Code	Integer	Postal code of the delivery location
Region	Text	Sales region (Central, East, South, West)
Product ID	Text	Identifier of the product
Category	Text	Product category (3 values)
Sub-Category	Text	Product sub-category (17 values)
Product Name	Text	Name of the product
Sales	Decimal	Sales amount of the order line
Quantity	Integer	Number of units sold
Discount	Decimal	Discount applied (0 to 0.80)
Profit	Decimal	Profit of the order line (negative for a loss)
Order Month	**Derived**	**Month number (1–12) taken from Order Date**
Order Year	**Derived**	**Year (2014–2017) taken from Order Date**
Order Weekday	**Derived**	**Day of the week (0 = Monday) taken from Order Date**

3.5 Design Selection: -

A notebook-based, library-driven design was selected: pandas for data handling, Seaborn and Matplotlib for charts, and Jupyter for combining code and results. Compared with a BI tool, it is free, transparent and reproducible. The chart type was chosen to match each question: a **line chart** for change over the months, a **donut chart** for the share of each category in the total, and a **horizontal bar chart** for the 17 sub-categories, because long labels and many bars are easier to read horizontally and negative profit values are clearly visible. Data labels were added to the bar charts so that the exact values can be read.

3.6 Implementation plan/methodology

Development was incremental: the data was loaded and checked first, then each analysis was written, run and visualised before the next one was started.

CHAPTER 4.
RESULTS ANALYSIS AND VALIDATION

4.1 Implementation of the solution:

The analysis was implemented in the Jupyter Notebook *ecommerce-analyis-jupyter.ipynb*. The sections below present the output of each module together with its interpretation. All amounts are in US dollars, as given in the dataset.

Data loading and profiling. The file was read with `pd.read_csv()` using `latin-1` encoding, giving 9,994 rows and 21 columns. The `info()` output showed that all columns are complete, and `isnull().sum()` and `duplicated().sum()` returned zero missing values and zero duplicate rows. Table 4.1 gives an overview of the data and Table 4.2 the statistical summary of the numeric columns.

Table 4.1: Overview of the dataset

Item	Value
Records (order lines) / columns	9,994 / 21 (24 after feature creation)
Distinct orders / customers / products	5,009 / 793 / 1,862
Period covered	3 January 2014 – 30 December 2017
Missing values / duplicate rows	0 / 0
Total sales	$2,297,200.86
Total profit	$286,397.02
Overall profit margin	12.47%

Table 4.2: Statistical summary of numeric columns

Column	Mean	Std. dev.	Minimum	Median	Maximum
Sales	229.86	623.25	0.44	54.49	22,638.48
Quantity	3.79	2.23	1.00	3.00	14.00
Discount	0.16	0.21	0.00	0.20	0.80
Profit	28.66	234.26	-6,599.98	8.67	8,399.98

The mean sales value ($229.86) is far above the median ($54.49) and the standard deviation is large, which shows that a few very large order lines pull the average up. Profit ranges from a loss of $6,599.98 to a gain of $8,399.98, so individual order lines can be strongly loss-making even though the total is positive.

Monthly sales and profit. Orders were grouped by the Order Month feature and the sales and profit were summed. Figure 3 shows that sales are lowest in February ($59,751) and then rise unevenly through the year, with the highest values in September ($307,650), November ($352,461) and December ($325,294). Taken together, October to December contribute 38.2% of all sales, and September to December contribute 51.6%. Note that the chart combines the same calendar month across all four years.

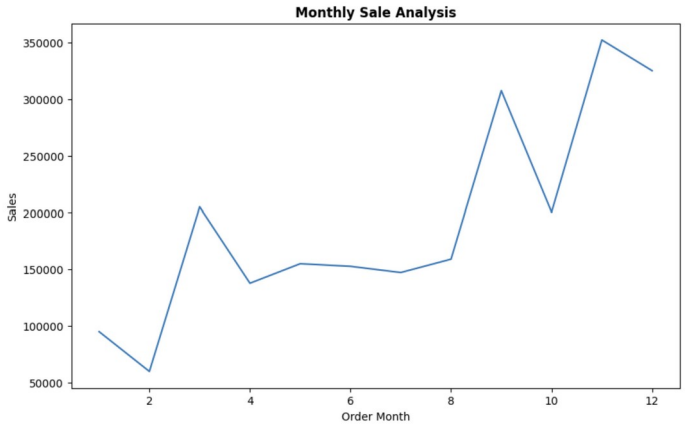


Figure 3: Monthly sales trend (all years combined)

Profit follows a similar upward pattern (Figure 4). The highest monthly profit is in December ($43,369) and the lowest in January ($9,134). The two charts do not peak together: November has the highest sales but only the fourth-highest profit, and its margin (10.1%) is below the yearly average. April (8.4%) and July (9.4%) have the lowest margins.

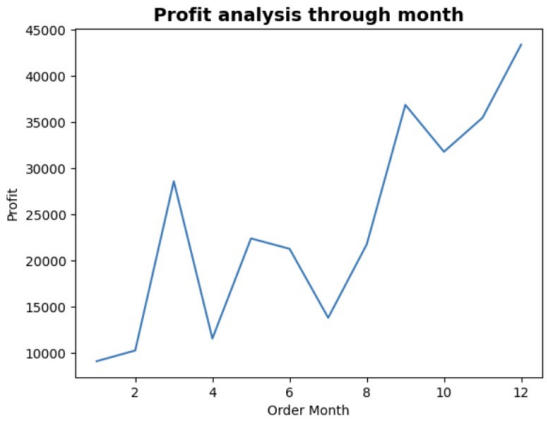


Figure 4: Monthly profit trend (all years combined)

Category analysis. Sales are spread almost evenly over the three categories, with Technology the largest at 36.4%, Furniture at 32.3% and Office Supplies at 31.3% (Figure 5). The picture is very different for profit (Figure 6): Technology gives 50.8% of the profit and Office Supplies 42.8%, whereas Furniture gives only 6.4% although it sells more than Office Supplies. Furniture's margin is 2.5%, against about 17% for the other two categories (Table 4.3).

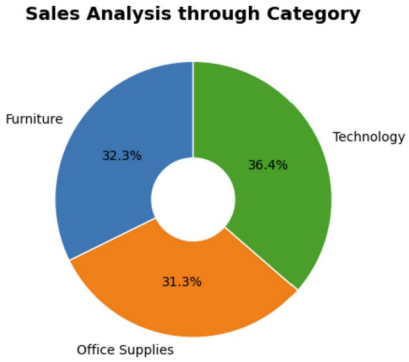


Figure 5: Sales share by product category

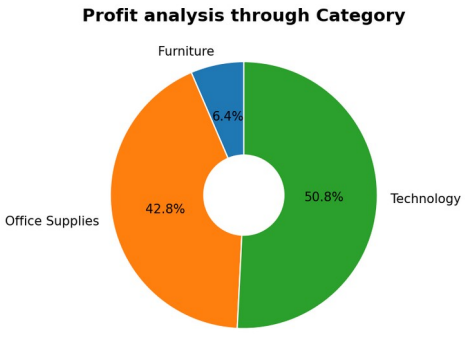


Figure 6: Profit share by product category

Table 4.3: Sales and profit by product category

Category	Sales ($)	Sales share	Profit ($)	Profit share	Margin
Furniture	741,999.80	32.3%	18,451.27	6.4%	2.5%
Office Supplies	719,047.03	31.3%	122,490.80	42.8%	17.0%
Technology	836,154.03	36.4%	145,454.95	50.8%	17.4%
Total	**2,297,200.86**	**100.0%**	**286,397.02**	**100.0%**	**12.5%**

Sub-category analysis. Figure 7 shows that Phones ($330,007) and Chairs ($328,449) are the two largest sub-categories by sales, followed by Storage, Tables and Binders, while Fasteners ($3,024), Labels ($12,486) and Envelopes ($16,476) are the smallest. The profit view (Figure 8) is quite different. Copiers earn the highest profit ($55,618) even though their sales are only mid-range, followed by Phones, Accessories and Paper. Three sub-categories make a loss: **Tables** (–$17,725), **Bookcases** (–$3,473) and **Supplies** (–$1,189).

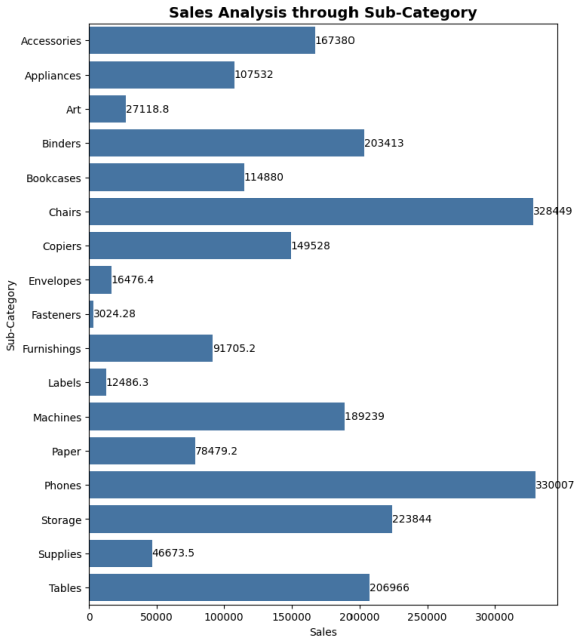


Figure 7: Sales by product sub-category

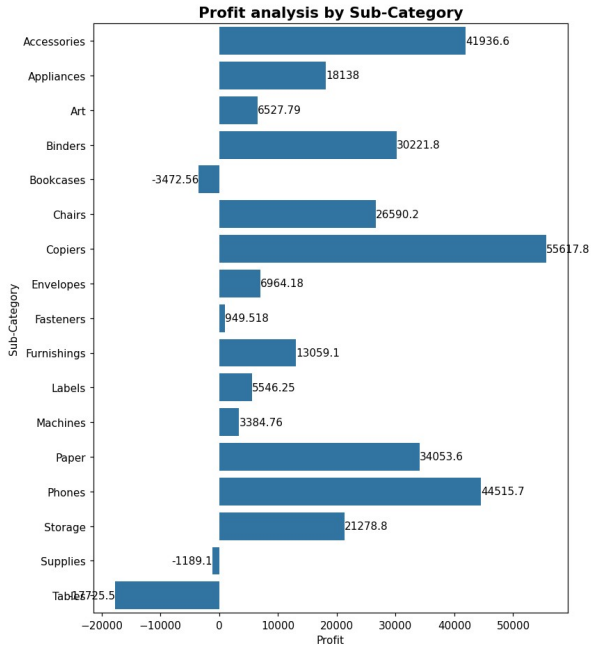


Figure 8: Profit by product sub-category

The table also shows that a high margin does not always mean a large contribution: Paper, Labels and Envelopes have margins above 40% but small sales, while Machines and Chairs have large sales but margins of only 1.8% and 8.1%.

Customer segment analysis. Figure 9 compares the three customer segments. The Consumer segment is the largest, with 50.6% of sales, followed by Corporate (30.7%) and Home Office (18.7%). The sales-to-profit ratio (Table 4.4) is lowest for Home Office (7.13) and highest for Consumer (8.66), which means that Home Office customers return the most profit per dollar of sales (margin 14.0%), and Consumer customers the least (11.5%).

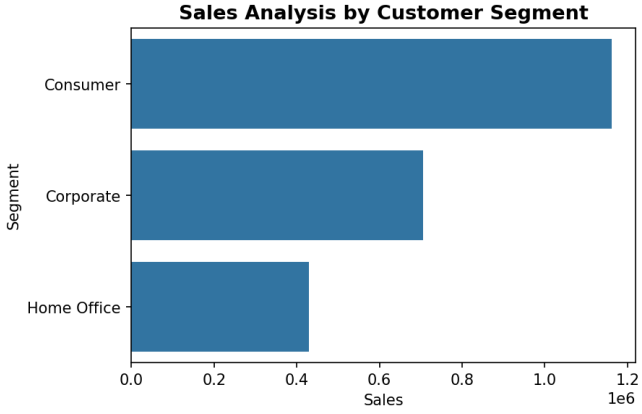


Figure 9: Sales by customer segment

Table 4.4: Sales, profit and sales-to-profit ratio by segment

Segment	Sales ($)	Sales share	Profit ($)	Margin	Sales : Profit
Consumer	1,161,401.35	50.6%	134,119.21	11.5%	8.66
Corporate	706,146.37	30.7%	91,979.13	13.0%	7.68
Home Office	429,653.15	18.7%	60,298.68	14.0%	7.13

4.2 Summary of findings and validation:

Key findings

- The business earned **$2,297,200.86** in sales and **$286,397.02** in profit over four years, an overall margin of **12.5%**.
- Sales are strongly seasonal: **September to December bring about 52% of annual sales**, and February is the weakest month.
- **Technology** is the largest and most profitable category (36.4% of sales, 50.8% of profit); **Furniture** is the weak spot (32.3% of sales, 6.4% of profit, 2.5% margin).
- **Copiers** are the most profitable sub-category, **Phones** and **Chairs** the best sellers; **Tables, Bookcases and Supplies** are loss-making.
- The **Consumer** segment gives half of the sales, but **Home Office** has the best margin and the lowest sales-to-profit ratio.

Validation

The figures in this report were checked against the dataset: the highest and lowest values were taken from the numeric Sales and Profit columns, and the grouped totals were reconciled with the overall totals. The checks are summarised below.

Table 4.5: Test summary

No.	Test case	Expected result	Result
1	Load the CSV with latin-1 encoding	9,994 rows × 21 columns loaded	Pass
2	Check for missing values	Zero missing values in every column	Pass
3	Check for duplicate rows	Zero duplicate rows	Pass
4	Convert Order Date and Ship Date	Both columns have date-time type	Pass
5	Create Order Month, Year and Weekday	Three new columns added	Pass
6	Reconcile monthly, category and sub-category sales	Each sums to the total sales of $2,297,200.86	Pass
7	Reconcile segment sales and profit	Totals equal overall sales and profit	Pass
8	Identify highest and lowest values	Taken from the numeric column with idxmax / idxmin	Pass
9	Draw all charts	Titles, axes and data labels displayed correctly	Pass

CHAPTER 5.
CONCLUSION AND FUTURE WORK

5.1 Conclusion: -

This project achieves its objectives: the Superstore dataset was loaded, checked and prepared with pandas, and its sales and profit were analysed by month, category, sub-category and customer segment and presented in clear charts. The analysis turned 9,994 raw order lines into a set of practical findings. Sales are concentrated in the last four months of the year; Technology and Office Supplies earn almost all of the profit; Furniture sells well but earns little; Tables, Bookcases and Supplies lose money; and Home Office customers are the most profitable per dollar of sales. The project also gave hands-on experience of the full life cycle of a data-analysis project – from data understanding and cleaning to visualisation, documentation and publication on GitHub.

Based on these findings, a retailer could (i) plan stock and staff for September to December; (ii) review the pricing and discounting of Tables and Bookcases; and (iii) give more attention to Technology and to high-margin lines such as Copiers and Paper. A quick additional check on the data supports the second point: more than half of the Tables order lines carry a discount of 30% or more, and those lines together lose about $30,700, while the remaining Tables lines make a profit. This is only a preliminary observation and should be examined properly in a discount analysis.

The present version is a foundation rather than a finished product. The monthly charts combine all four years, so year-to-year growth is not shown; the charts are static; the analysis is descriptive and contains no forecasting or statistical testing; and the dataset is a public sample, so the findings illustrate the method rather than describe a real company. These limitations form the basis of the future scope.

5.2 Future Scope: -

- Discount, region and state analysis to explain the causes of losses, including the effect of discount levels on profit.
- Year-over-year analysis (the Order Year feature is already available) and sales forecasting with time-series models.
- Customer-level analysis: RFM scoring, customer lifetime value and clustering (e.g. K-Means) to find high-value customers.
- Shipping analysis: the relation between ship mode, delivery time and profit.
- An interactive dashboard in Power BI, Tableau, Streamlit or Plotly so that managers can filter by period, region or category.
- Packaging the analysis as a reusable script or pipeline with automated checks, and connecting it to a database instead of a CSV file.

APPLICATION CODE OF PROJECT

The main code cells of the notebook are given below. The complete notebook, *ecommerce-analyis-jupyter.ipynb*, and the dataset are available in the GitHub repository. In the notebook the weekday column is named "Order day of the month"; it stores the day of the week and is shown here as Order Weekday for clarity.

1. Importing libraries and loading the data

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

data = pd.read_csv("Sample - Superstore.csv", encoding='latin-1')
data.info()
data.describe()
```

2. Data preparation and feature engineering

```
# check data quality
data.isnull().sum()
data.duplicated().sum()

# convert dates to date-time type
data['Order Date'] = pd.to_datetime(data['Order Date'])
data['Ship Date']  = pd.to_datetime(data['Ship Date'])

# derive time features
data['Order Month']   = data['Order Date'].dt.month
data['Order Year']    = data['Order Date'].dt.year
data['Order Weekday'] = data['Order Date'].dt.dayofweek   # 0 = Monday
```

REFERENCES

1. Tukey, J. W. (1977). Exploratory Data Analysis. Addison-Wesley.
2. McKinney, W. (2022). Python for Data Analysis (3rd ed.). O'Reilly Media.
3. VanderPlas, J. (2016). Python Data Science Handbook. O'Reilly Media.
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5. Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. Computing in Science & Engineering, 9(3), 90–95.
6. Waskom, M. L. (2021). seaborn: statistical data visualization. Journal of Open Source Software, 6(60), 3021.
7. The pandas development team. pandas documentation. https://pandas.pydata.org/docs/

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9. Tableau Software. Sample – Superstore dataset (sample data distributed with Tableau Desktop).

USER MANUAL

A. Setting up the project

1. Install Python 3 and the required libraries: `pip install pandas numpy matplotlib seaborn notebook`.
2. Download or clone the GitHub repository and keep `Sample - Superstore.csv` in the same folder as the notebook.
3. Start Jupyter with the command `jupyter notebook` and open `ecommerce-analyis-jupyter.ipynb`.

B. Using the notebook

1. Run the cells from top to bottom (Kernel → Restart & Run All). The data is loaded and the summary statistics are shown.
2. Check the data-quality cells: both the missing-value and duplicate checks should return zero.
3. Read the monthly sales and profit charts (Figures 3 and 4) to see the seasonal pattern.
4. Read the category and sub-category charts (Figures 5 to 8) to see which products sell and which earn profit.
5. Read the segment chart and the sales-to-profit ratio (Figure 9) to compare customer groups.
6. To analyse another sales file, replace the CSV with a file that has the same column names and run the notebook again.

GITHUB REPOSITORY

The complete project (notebook, dataset, README and chart images) is published on GitHub at: https://github.com/jindalbhumi125-dev/E-Commerce-Sales-and-Customer-Intelligence

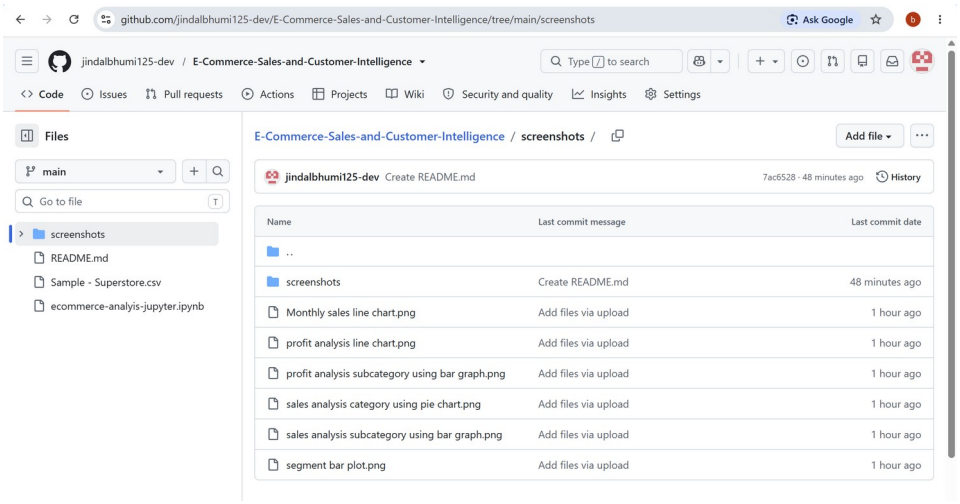


Figure 10: GitHub repository of the project